

Extended information

Machine learning-based identification of key biotic and abiotic drivers of mineral weathering rate in a complex enhanced weathering experiment

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A Materials

A.1 Irrigated water

Table A1: Ions measured in the natural groundwater used for irrigation of the microcosms (mean $\pm$ standard deviation, $n = 10$). Values below the detection limit are indicated as '< limit'.

Property	Amount	Unit
Ca^{2+}	714.72 ± 14.6	mmol l^{-1}
K^{+}	23.59 ± 9.02	mmol l^{-1}
Mg^{2+}	105.42 ± 17.68	mmol l^{-1}
P	0.29 ± 0.25	mmol l^{-1}
Na^{+}	283.22 ± 11.60	mmol l^{-1}
F^{-}	6.19 ± 3.38	mmol l^{-1}
Cl^{-}	237.89 ± 23.65	mmol l^{-1}
NO_2^{-}	< 0.01	mmol l^{-1}
SO_4^{2-}	98.09 ± 11.32	mmol l^{-1}
PO_4^{2-3-}	< 0.01	mmol l^{-1}
Br^{-}	< 0.005	mmol l^{-1}

Table A2: Properties of the natural groundwater (mean $\pm$ standard deviation, $n = 6$).

Property	Amount	Unit
pH	7.85 ± 0.41	
Electrical conductivity	198.33 ± 8.68	$\mu\text{S cm}^{-1}$
Dissolved silica	222.23 ± 3.69	$\mu\text{mol l}^{-1}$
Temperature	12.53 ± 3.35	$^{\circ}\text{C}$
Total alkalinity	1548.95 ± 29.55	$\mu\text{mol l}^{-1}$
Dissolved inorganic carbon	12.94 ± 0.30	mg l^{-1}
Dissolved organic carbon	4.98 ± 0.59	mg l^{-1}

A.2 Rocks and minerals

Table A3: Sources and references of the used rocks and slags

Rock powder	Source	References
Basalt	<i>RPBL Basalt and Lava aus der Eifel</i> , Germany	Jansen et al. [2024]
Basanite/tephrite	<i>RPBL Basalt and Lava aus der Eifel</i> , Germany	Sundermeyer et al. [2020], Jansen et al. [2024]
Peridotite	Åheim, <i>Sibelco</i> , Norway	Kremer et al. [2019], Wang et al. [2019]
Steel slag	<i>ThyssenKrupp</i> , Germany	

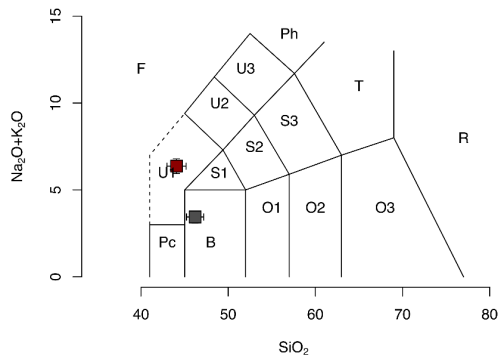


Figure A1: Total Alkali Silica diagram to classify the basaltic rocks in this study. The grey square indicates basalt, which is classified as basalt rock; the red square denotes basanite/tephrite, which falls in the "U1" and is classified as basanite/tephrite.

Table A4: Qualitative composition of the rock powders determined via X-ray diffraction (XRD). Asterisks indicate the minerals' fraction that were not detected via CIPW norm approach.

Rock powders	Mineral composition
Basalt	Augite*
	Anorthite
	Albite
basanite/tephrite	Diopside
	Phlogopite*
	Albite*
	Nepheline*
Peridotite	Forsterite
	Clinocllore*
	Chabazite
	Antigorite*
Steel slag	Magnetite
	Portlandite
	Larnite
	Srebrodolskite
	Wustite

Table A5: Major contributions to the normative composition of peridotite, basalt and basanite/tephrite, according to their elemental compositions calculated using the CIPW norm approach from Cross et al. [1902]. Asterisks indicate the minerals' fraction that was not detected with X-ray diffraction (XRD).

Normative Minerals	Peridotite [%]	Basalt [%]	Basanite/tephrite [%]
Corundum	0.082		
Orthoclase	0.183	6.277	4.107
Albite	0.261	18.862*	
Anorthite	1.724	21.032*	14.772
Apatite	0.023	1.515	1.249
Nepheline		0.626	13.753
Leucite			12.222
Zircon	0.002	0.042	0.053
Diopside		23.28	36.159*
Enstatite	11.949		
Hypersthene		0.539	
Forsterite	76.907*	14.403	5.472
Fayalite	3.827	2.966	1.315
Magnetite	4.332	6.615	5.814
Ilmenite	0.038	4.289	4.955
Thenardite		0.018	0.035
Chromite	0.133	0.077	
Sum	100	100.002	99.999
Fe ratio (Le Maitre)	0.75	0.66	0.61

Table A6: Composition of the rock powders as determined by X-ray fluorescence (mean $\pm$ standard deviation, n = 3). B.D.L. = below detection limit.

Element	Peridotite [%]	Basalt [%]	Steel slag [%]	Basanite/tephrite [%]
SiO ₂	41.55 $\pm$ 0.23	43.87 $\pm$ 0.55	11.26 $\pm$ 0.01	43.44 $\pm$ 0.27
TiO ₂	0.02 $\pm$ 0.01	2.15 $\pm$ 0.06	0.52 $\pm$ 0.04	2.58 $\pm$ 0.33
Al ₂ O ₃	0.78 $\pm$ 0.01	12.13 $\pm$ 0.18	3.21 $\pm$ 0.65	13.79 $\pm$ 0.25
Fe ₂ O ₃	7.26 $\pm$ 0.01	11.66 $\pm$ 0.18	22.69 $\pm$ 3.89	11.19 $\pm$ 0.26
MnO	0.09 $\pm$ 0.01	0.17 $\pm$ 0.01	1.46 $\pm$ 0.05	0.17 $\pm$ 0.01
MgO	47.87 $\pm$ 0.24	11.4 $\pm$ 0.31	4.7 $\pm$ 0.25	8.68 $\pm$ 0.12
CaO	0.35 $\pm$ 0.19	10.41 $\pm$ 0.02	41.95 $\pm$ 0.63	12.6 $\pm$ 0.1
Na ₂ O	0.03 $\pm$ 0.01	2.26 $\pm$ 0.01	B.D.L.	2.99 $\pm$ 0.08
K ₂ O	0.03 $\pm$ 0.01	1.01 $\pm$ 0.06	B.D.L.	3.29 $\pm$ 0.09
P ₂ O ₅	0.01 $\pm$ 0.01	0.62 $\pm$ 0.01	1.32 $\pm$ 0.01	0.53 $\pm$ 0.07
SO ₃	B.D.L.	0.01 $\pm$ 1.00	0.10 $\pm$ 0.03	0.03 $\pm$ 0.01
LOI	1.8 $\pm$ 0.13	4.2 $\pm$ 2.00	12.59 $\pm$ 4.55	0.62 $\pm$ 0.08
Sum	99.78 $\pm$ 0.12	99.91 $\pm$ 0.07	99.8 $\pm$ 0.83	99.91 $\pm$ 0.44

Table A7: Composition of the rock powders as determined by X-ray fluorescence (mean $\pm$ standard deviation, n = 3). B.D.L. = below detection limit.

Element	Peridotite [ppm]	Basalt [ppm]	Steel Slag [ppm]	Basanite/tephrite [ppm]
As	18.05 $\pm$ 7.79	8.83 $\pm$ 4.44	25.61 $\pm$ 11.56	26.25 $\pm$ 9.86
Ba	17.59 $\pm$ 2.87	542.8 $\pm$ 27.08	66.35 $\pm$ 4.56	9045 $\pm$ 88.03
Bi	1.7 $\pm$ 2.38	4.17 $\pm$ 0.71	3.94 $\pm$ 0.25	2.58 $\pm$ 3.65
Ce	8.08 $\pm$ 0.12	93.19 $\pm$ 7.02	B.D.L.	130.04 $\pm$ 27.81
Co	122.38 $\pm$ 12.68	62.86 $\pm$ 15.66	33.07 $\pm$ 30.46	86.65 $\pm$ 5.2
Cr	600.96 $\pm$ 22.01	141.81 $\pm$ 13.87	273.63 $\pm$ 13.14	73.03 $\pm$ 11.27
Cu	B.D.L.	63.96 $\pm$ 1.58	B.D.L.	83.38 $\pm$ 1.21
Ga	2.92 $\pm$ 0.11	13.35 $\pm$ 2.17	4.21 $\pm$ 0.51	14.98 $\pm$ 1.47
La	7.12 $\pm$ 8.31	0.45 $\pm$ 0.82	0.569 $\pm$ 0.542	0.503 $\pm$ 1.223
Nb	0.1289 $\pm$ 0.538	0.5275 $\pm$ 0.695	0.13576 $\pm$ 0.502	0.5529 $\pm$ 0.164
Nd	B.D.L.	0.3541 $\pm$ 0.436	B.D.L.	0.4502 $\pm$ 0.883
Ni	247.814 $\pm$ 5.94	335.24 $\pm$ 20.24	B.D.L.	114.23 $\pm$ 11.68
Pb	3.06 $\pm$ 2.45	5.16 $\pm$ 0.37	11.94 $\pm$ 0.08	4.73 $\pm$ 2.27
Rb	1.73 $\pm$ 2.03	32.84 $\pm$ 0.02	B.D.L.	77.01 $\pm$ 2.75
Sc	6.2 $\pm$ 0.28	24.83 $\pm$ 4.03	B.D.L.	38.76 $\pm$ 19.23
Sr	29.11 $\pm$ 2.77	7348 $\pm$ 13.17	128.04 $\pm$ 3.55	836.62 $\pm$ 44.74
Th	3.25 $\pm$ 0.35	4.18 $\pm$ 1.14	4.31 $\pm$ 0.03	6.59 $\pm$ 3.81
U	0.28 $\pm$ 0.33	5.41 $\pm$ 0.57	1.53 $\pm$ 0.05	2.21 $\pm$ 1.71
V	20.67 $\pm$ 7.44	213.06 $\pm$ 8.07	1429.42 $\pm$ 11	273.77 $\pm$ 18.07
Y	2.17 $\pm$ 0.23	24.92 $\pm$ 0.31	B.D.L.	25.15 $\pm$ 1.47
Zn	37.72 $\pm$ 0.99	99.99 $\pm$ 4.15	21 $\pm$ 7.78	80.19 $\pm$ 0.37
Zr	9.59 $\pm$ 3.39	208.37 $\pm$ 4.58	110.03 $\pm$ 14.28	264.94 $\pm$ 6.84

A.3 Biochar and organic amendments

Table A8: Properties of the used biochar, as reported by *Eurofins*. The bulk density was measured on material sieved to below 3 mm; the specific surface area was determined using the Brunauer–Emmett–Teller theory; and the water holding capacity was determined on material sieved to below 2 mm.

Property	amount
Bulk density	207 kg m ³
Specific surface area	314.89 m ² g ⁻¹
Water holding capacity	308.1 %
Ash content	15.3 mass%
Total carbon	80.5 mass%
Total organic carbon	79.6 mass%
pH in CaCl ₂	8.6

Table A9: Elemental composition (mean $\pm$ standard deviation, $n = 4$) of the used biochar, straw, digestate and alfalfa mixture. Total carbon (C) and nitrogen (N) contents were determined using an elemental analyser (FlashSmart, Thermo Fisher Scientific, USA). The other elements were determined using ICP-OES (Thermo iCAP 6500 dual view) after digestion with nitric- and hydrochloric acid and heating in a microwave, adapted from Novozamsky et al. [1996].

Element	Unit	Straw	Digestate	Alfalfa mixture	Biochar
Al	[mg/kg]	17.3 $\pm$ 1.9	187 $\pm$ 41	97.8 $\pm$ 9.6	244.3 $\pm$ 9.7
Ca	[mg/g]	2.915 $\pm$ 0.071	12.39 $\pm$ 0.92	9.60 $\pm$ 0.35	19.6 $\pm$ 1.3
Cu	[mg/kg]	1.950 $\pm$ 0.058	38.48 $\pm$ 0.74	5.78 $\pm$ 0.15	14.4 $\pm$ 1.3
Fe	[mg/g]	0.112 $\pm$ 0.017	1.176 $\pm$ 0.093	0.225 $\pm$ 0.025	0.326 $\pm$ 0.033
K	[mg/g]	8.000 $\pm$ 0.082	2.750 $\pm$ 0.058	17.93 $\pm$ 0.28	6.575 $\pm$ 0.050
Mg	[mg/g]	0.4575 $\pm$ 0.0096	1.313 $\pm$ 0.034	1.320 $\pm$ 0.025	1.53 $\pm$ 0.20
Mn	[mg/kg]	50.75 $\pm$ 0.96	119.8 $\pm$ 5.2	25.25 $\pm$ 0.50	204 $\pm$ 11
Na	[mg/kg]	19.0 $\pm$ 2.0	924 $\pm$ 11	299.0 $\pm$ 5.5	50.5 $\pm$ 1.3
P	[mg/g]	0.4725 $\pm$ 0.0096	4.72 $\pm$ 0.38	2.368 $\pm$ 0.057	1.410 $\pm$ 0.025
Zn	[mg/kg]	4.25 $\pm$ 0.50	144.0 $\pm$ 6.4	18.75 $\pm$ 0.50	89.0 $\pm$ 1.9
C	[mg/g]	5.15 $\pm$ 0.99	8.78 $\pm$ 0.54	21.00 $\pm$ 0.22	5.53 $\pm$ 0.27
N	[mg/g]	443.7 $\pm$ 9.9	442.2 $\pm$ 6.9	430.4 $\pm$ 1.5	782.8 $\pm$ 7.3

A.4 Alkalinity, dissolved and solid inorganic carbon of the materials

Table A10: Concentrations of dissolved inorganic carbon (DIC), dissolved organic carbon (DOC) and total alkalinity (TA), measured in leached water of blank columns, with a solid bed consisting only of sand, biochar and organic amendments.

	Solid bed	V_{irr} [l]	DIC [$\frac{mg}{l}$]	DOC [$\frac{mg}{l}$]	TA [$\frac{\mu mol}{l}$]
B_1	400g sand	5.6	11.875	7.275	1434.9
B_2	400g sand	5.6	11.088	5.120	1354.5
B_3	400g sand	8.4	11.783	6.102	1427.3
$B_{straw,1}$	400g sand + 10g straw	5.6	16.408	35.169	1981.1
$B_{straw,2}$	400g sand + 10g straw	8.4	13.589	27.364	1632.9
$B_{digestate,1}$	400g sand + 10g digestate	5.6	10.502	16.942	1445.4
$B_{alfalfa,1}$	400g sand + 10g alfalfa mixture	5.6	26.387	44.609	2304.8
$B_{biochar,1}$	400g sand + 20g biochar	5.6	15.452	5.561	1761.3
$B_{biochar,2}$	400g sand + 40g biochar	5.6	18.387	6.330	2053.5
$B_{biochar,3}$	400g sand + 20g biochar	8.4	14.708	5.482	1698.6

Table A11: Specific surface area measurements of rock/slag flours, loss of ignition and solid inorganic carbon measurements of samples of rock flours, biochar and organic amendments. SSA = Specific surface area, LOI = loss of ignition, SIC = solid inorganic carbon. SSA measurements of rocks are obtained using Brunauer-Emmett-Teller method. $m_{\text{solid sample}}^{\text{pre-LOI}}$ and $m_{\text{solid sample}}^{\text{post-LOI}}$ are the gross sample masses before and after loss of ignition (LOI), m_{crucible} is the mass of the crucible holding the sample and SIC is the measured solid inorganic carbon concentration.

Matter		SSA [m ² /g]	$m_{\text{sample},X}^{\text{pre-LOI}}$ [g]	$m_{\text{sample},X}^{\text{post-LOI}}$ [g]	$m_{\text{crucible},X}$ [g]	$\text{SIC}_{\text{conc}}^X$ [mass%]
Basalt	≤ 0.063 mm	1.896	37.5	37.4	20.9	0.009
	0.063-2 mm	0.513	41.9	41.9	20.2	0.006
Basanite/tephrite	≤ 0.063 mm	10.577	38.4	38.3	19.7	0.134
	0.063-3 mm	6.972	40.2	40.0	21.3	0.068
Peridotite	≤ 0.063 mm	7.050	36.2	35.8	18.8	0.031
	0.063-0.125 mm	0.959	45.2	45.1	19.8	0.005
	0.125-1 mm	1.134	51.2	51.1	20.0	0.008
Steelslag	≤ 0.1 mm	13.246	32.6	31.8	18.8	1.922
	0.1-0.5 mm	9.154	39.9	38.9	19.3	1.858
	0.5-3 mm	4.747	38.8	38.3	18.8	1.359
	mixed (≤ 3 mm)	10.271	29.6	29.2	19.0	0.992
Digestate		-	22.2	20.2	20.0	2.272
Straw		-	21.0	20.0	20.0	4.957
Alfalfa mixture		-	23.7	23.3	23.3	3.339
Biochar		-	23.8	20.5	19.5	0.263

B Data processing

B.1 Calculation of change in amount of dissolved inorganic carbon, solid inorganic carbon, total inorganic carbon and total alkalinity

This section outlines the methodology used to calculate the changes in the amount of dissolved inorganic carbon (DIC), solid inorganic carbon (SIC), total inorganic carbon (TIC), and total alkalinity (TA). Detailed equations and relevant measurements are provided to ensure clarity and reproducibility.

B.1.1 Change in amount of dissolved inorganic carbon

The change in amount of DIC (ΔDIC) is defined as the difference between the final (DIC_f) and the initial DIC present at the beginning of the experiment (DIC_0):

$$\Delta DIC [mmol] = DIC_f [mmol] - DIC_0 [mmol] \quad (B.1)$$

Here, DIC_f is calculated using the DIC concentration measured after 8 weeks (DIC_{conc}) and the irrigated volume (V_{irr}):

$$DIC_f [mmol] = DIC_{conc} \left[\frac{mg}{l} \right] \cdot V_{irr} [l] \cdot \frac{1 [mmol]}{12.001 [mg]} \quad (B.2)$$

The DIC_0 is the sum of the DIC present in the irrigated water and the DIC stemming from the organic amendments and biochar. To calculate the DIC stemming from the irrigated water, the DIC concentration of columns with a solid bed consisting only of 400g sand (coined B), was measured (DIC_{conc}^B) and averaged (Tab. A10).

$$DIC_{conc}^{water} \left[\frac{mg}{l} \right] = \overline{DIC_{conc}^{B_i} \left[\frac{mg}{l} \right]} \quad (B.3)$$

The DIC stemming from the organic amendments and biochar was based on DIC concentration measurements of columns with a solid bed made out of only sand and straw (B_{straw}), digestate ($B_{digestate}$), alfalfa-dominated mixture ($B_{alfalfa}$) or biochar ($B_{biochar}$) (Tab. A10):

$$DIC_{conc}^{matter} \left[\frac{mg}{g \text{ matter}} \right] = \frac{(\overline{DIC_{conc}^{B_{matter,i}} \left[\frac{mg}{l} \right]} - DIC_{conc}^{water} \left[\frac{mg}{l} \right]) \cdot V_{irr}^{B_{matter,i}} [l]}{m_{org,i}^{B_{matter,i}} [g]} \quad (B.4)$$

Here, *matter* can be either straw, digestate, the alfalfa mixture and biochar.

Finally, the initial DIC was calculated using:

$$DIC_0 [mmol] = \left(DIC_{conc}^{water} [mg] \cdot V_{irr} [l] + \sum_{matter} DIC_{conc}^{matter} \left[\frac{mg}{g \text{ matter}} \right] \cdot m_{matter} [g] \right) \cdot \frac{1 [mmol]}{12.001 [mg]} \quad (B.5)$$

Here, the summation goes over straw, digestate, the alfalfa mixture and biochar.

B.1.2 Change in amount of solid inorganic carbon

Analogously to ΔDIC , change in amount of SIC (ΔSIC) is the difference between the final (SIC_f) and the initial SIC that is added to the system through the input materials (SIC_0):

$$\Delta SIC [mmol] = SIC_f [mmol] - SIC_0 [mmol] \quad (B.6)$$

Before measuring the SIC concentration (SIC_{conc}) of the solid samples (using *C analysis, Flash 2000 CN Soil Analyser, Interscience, Antwerp, Belgium*), LOI is performed, which lead to mass loss. Therefore, SIC_f is calculated using:

$$SIC_f [mmol] = 10 \cdot SIC_{conc} [mass\%] \cdot LOI \text{ correction} \cdot m_{solid} [g] \cdot \frac{1 [mmol]}{12.001 [mg]} \quad (B.7)$$

where m_{solid} is the mass of the added rocks, organic amendments and biochar; and the *LOI correction* factor accounts for sample mass loss during LOI:

$$LOI\ correction = 1 - \frac{m_{solid\ sample}^{post-LOI} [g] - m_{crucible} [g]}{m_{solid\ sample}^{pre-LOI} [g] - m_{crucible} [g]} \quad (B.8)$$

with $m_{solid\ sample}^{pre-LOI}$ and $m_{solid\ sample}^{post-LOI}$ the gross sample mass before and after LOI, and $m_{crucible}$ the mass of the crucible holding the solid sample.

SIC_0 is the *SIC* present at the start of the experiment and is calculated based on *SIC* measurements of the added rocks, organic amendments and biochar (Tab. A11) and their masses in the column ($m_{sample,X}$):

$$SIC_0 [mmol] = 10 \cdot \sum_X SIC_{conc}^X [mass\%] \cdot LOI\ correction_X \cdot m_{sample,X} [g] \cdot \frac{1 [mmol]}{12.001 [mg]} \quad (B.9)$$

where the summation goes over the added rocks, organic amendments and biochar; and the *LOI correction* factor again accounts for sample mass loss during LOI (Eq. B.8).

B.1.3 Change in amount of total inorganic carbon

In this study, total inorganic carbon is defined as the sum of solid inorganic carbon and dissolved inorganic carbon. Therefore, the final TIC amount, initial TIC amount and change in amount of TIC (ΔTIC) are calculated as:

$$TIC_f [mmol] = SIC_f [mmol] + DIC_f [mmol] \quad (B.10)$$

$$TIC_0 [mmol] = SIC_0 [mmol] + DIC_0 [mmol] \quad (B.11)$$

$$\Delta TIC [mmol] = \Delta SIC [mmol] + \Delta DIC [mmol] \quad (B.12)$$

B.1.4 Change in amount of total alkalinity

The change in amount of TA (ΔTA) is the difference between the final *TA* amount (TA_f) and the initial *TA* amount that results from non-weathering processes (TA_0):

$$\Delta TA [mmol] = TA_f [mmol] - TA_0 [mmol] \quad (B.13)$$

The final *TA* is calculated from the measured *TA* concentration at the end of the experiment (TA_{conc}) and the irrigated volume:

$$TA_f [mmol] = 10^{-3} TA_{conc} \left[\frac{\mu mol}{l} \right] \cdot V_{irr} [l] \quad (B.14)$$

The initial *TA* is calculated analogously to the initial *DIC* (Equations B.3-B.5), i.e., based on *TA* concentration measurements of batches with a solid bed consisting only of 400g sand (*B*) and batches with a solid bed made out of only organic amendments and biochar (B_{straw} , $B_{digestate}$, $B_{alfalfa}$, $B_{biochar}$) (Tab. A10):

$$TA_{conc}^{water} \left[\frac{mmol}{l} \right] = 10^{-3} \overline{TA_{conc}^{B_i} \left[\frac{\mu mol}{l} \right]} \quad (B.15)$$

$$TA_{conc}^{matter} \left[\frac{mmol}{g\ matter} \right] = 10^{-3} \frac{(TA_{conc}^{B_{matter,i}} \left[\frac{\mu mol}{l} \right] - TA_{conc}^{water} \left[\frac{\mu mol}{l} \right]) \cdot V_{irr}^{B_{matter,i}} [l]}{m_{matter,i}^{B_{matter,i}} [g]} \quad (B.16)$$

$$TA_0 [mmol] = TA_{conc}^{water} \left[\frac{mmol}{l} \right] \cdot V_{irr} [l] + \sum_{matter} TA_{conc}^{matter} \left[\frac{mmol}{g\ matter} \right] \cdot m_{matter} [g] \quad (B.17)$$

where the summation goes over straw, digestate, the alfalfa mixture and biochar.

B.2 Transformation of model features

Some features were transformed before being used as input for the machine learning (ML) models. The relevant equations for such features are provided below.

The total surface area of rock type M is calculated using:

$$SA_M [m^2] = \sum_{G}^{grain\ sizes} mass_{MG} [g] \cdot SSA_{MG} \left[\frac{m^2}{g} \right] \quad (B.18)$$

where $mass_{MG}$ is the added mass, and SSA_{MG} is the specific surface area of rock type M with grain size G (Table A11). The feature *chlorotica ratio* is defined as the ratio of initially live *A. chlorotica* earthworms with respect to the total number of initially live earthworms (i.e., the sum of the number of *A. chlorotica* earthworms and the number of *A. caliginosa* earthworms):

$$chlorotica\ ratio = \frac{number\ of\ initially\ live\ A.\ chlorotica\ earthworms}{number\ of\ initially\ live\ earthworms} \quad (B.19)$$

And analogously, the feature $\dagger$ *chlorotica ratio* is defined as:

$$\dagger\ chlorotica\ ratio = \frac{number\ of\ initially\ dead\ A.\ chlorotica\ earthworms}{number\ of\ initially\ dead\ earthworms} \quad (B.20)$$

B.3 Analysis of noise in data

To quantify the noise in the data, some of the input combinations of the columns was repeated. This resulted in 28 combinations that were repeated at least three times for ΔDIC , 30 for ΔSIC , 27 for ΔTIC and 27 for ΔTA . For each set of repeats, the standard deviation of the weathering indicators (ΔDIC , ΔSIC , ΔTIC and acDTA) was calculated and rescaled by dividing by the interquartile range of the indicator of the full dataset (Fig. B1). As steel slag generally lead to different scales of the indicators, this rescaling was done separately for the subset of columns with and without steel slag. For all indicators, the resulting rescaled standard deviation was relatively high, especially for ΔSIC and, to a lesser extent, ΔTIC (Fig. B1).

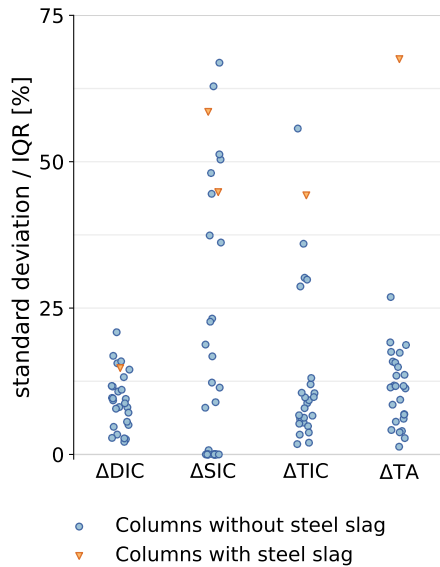


Figure B1: Quantification of the noise in data of the changes of dissolved inorganic carbon (ΔDIC), solid inorganic carbon (ΔSIC), total inorganic carbon (ΔTIC) and total alkalinity (ΔTA). The y axis depicts the standard deviations of columns that were repeated at least three times, rescaled by interquartile range (IQR) of the full dataset. The rescaling was done separately for the datasets with (orange) and without steel slag (blue).

To assess whether this noise resulted from certain inputs of the columns, we used a Random Forest (RF) to predict the standard deviations of the target values of repeated combinations for each of the indicators (ΔDIC , ΔSIC , ΔTIC , ΔTA). As the number of repeated columns was limited, leave one out cross validation (CV) was used. However, the model's performances were poor for all indicators, except for the prediction of ΔSIC (Tab. B1). A subsequent Shapley

Additive exPlanations (SHAP) analysis revealed that high standard deviations were predicted for columns with steel slag (B2). Therefore, the prediction of ΔSIC was repeated for columns without steel slag, which resulted in poor performances (Tab. B1). This suggests that, while steel slag leads to noisy measurements for SIC , the remaining noise of the indicators likely arises from other factors than the column inputs, such as sampling inconsistencies from imperfectly homogenized columns or irregular irrigation.

		MAE	MSE	RMSE	R^2
ΔDIC	RF	0.37	0.45	0.20	-0.50
	naïve	0.30	0.38	0.14	-0.08
ΔSIC	RF	3.2	8.4	70	0.73
	naïve	8.2	16.5	272	-0.07
ΔTIC	RF	2.8	7.9	62	-0.03
	naïve	3.5	8.1	65	-0.08
ΔTA	RF	1.6	5.1	26	-0.20
	naïve	1.8	4.8	23	-0.08
ΔSIC (without steel slag)	RF	0.97	1.2	1.4	0.23
	naïve	1.21	1.4	2.0	-0.08

Table B1: Performance of the Random Forest prediction of the standard deviation of ΔDIC , ΔSIC , ΔTIC and ΔTA of repeated measurements; and of ΔSIC of repeated measurements without steel slag. Depicted performance metrics are mean absolute error (MAE), mean squared error (MSE), root mean squared error (RMSE), and coefficient of determination (R^2). The performance of the naïve regressor, which predicts the mean target value for the whole dataset, is added as a baseline.

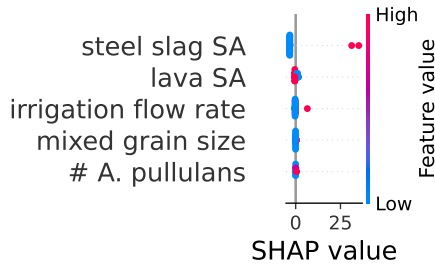


Figure B2: SHapley Additive exPlanation (SHAP) values of the top 5 most important features in the random forest prediction of the standard deviation of the change in amount of solid inorganic carbon of repeated measurements.

C Machine learning analysis

C.1 Tuned hyperparameters

Table C1: Tuned and selected hyperparameters in each fold of the outer loop of the nested cross validation, for the Least absolute shrinkage and selection operator regularization (Lasso), Random Forest (RF) and eXtreme Gradient Boosting (XGB) models; as defined in the scikit-learn [Pedregosa et al., 2011] and XGBoost [Chen and Guestrin, 2016] libraries.

Model	Tuned hyperparameter	Possible values	Target	Selected values per fold									
				1	2	3	4	5	6	7	8	9	10
Lasso	alpha	[0.001, 10]	ΔDIC	0.006	0.017	0.002	0.006	0.004	0.006	0.009	0.003	0.009	0.006
			ΔSIC	0.050	0.050	0.100	0.071	0.071	0.035	0.071	0.025	0.035	0.100
			ΔTIC	0.025	0.050	0.050	0.071	0.035	0.035	0.050	0.050	0.050	0.035
			ΔTA	0.071	0.100	0.071	0.100	0.071	0.0071	0.071	0.050	0.100	0.100
RF	max_depth	{3, 4, 5}	ΔDIC	5	5	5	5	5	5	5	5	5	5
			ΔSIC	5	5	5	5	5	5	5	5	5	5
			ΔTIC	5	5	5	5	5	5	5	5	5	5
			ΔTA	5	5	5	5	5	5	5	5	4	5
	n_estimators	{25, 50, 100, 200, 400}	ΔDIC	25	400	50	25	400	200	50	100	100	400
			ΔSIC	100	200	25	200	50	25	200	100	400	25
			ΔTIC	25	400	25	400	25	50	200	25	200	400
			ΔTA	400	25	200	100	400	200	200	100	25	50
	min_samples_split	{6, 8, 10}	ΔDIC	8	6	6	8	10	10	6	10	6	10
			ΔSIC	6	10	8	6	6	6	6	8	6	8
			ΔTIC	6	6	6	6	6	6	6	6	6	6
			ΔTA	6	6	6	10	6	6	6	6	6	6
			ΔDIC	3	3	4	5	3	3	4	5	3	5
			ΔSIC	5	4	4	5	3	5	5	5	5	5
			ΔTIC	4	4	5	4	4	5	3	3	4	5
			ΔTA	3	5	3	4	3	5	3	3	3	3
	n_estimators	{25, 50, 100}	ΔDIC	100	100	100	100	100	100	100	100	100	50
			ΔSIC	50	25	25	50	25	25	25	25	25	25
			ΔTIC	100	100	100	100	100	100	100	100	100	100
			ΔTA	25	50	100	25	100	25	50	100	25	100
	min_child_weight	{7, 8, 9, 10}	ΔDIC	9	9	8	7	9	7	8	10	7	10
			ΔSIC	10	10	10	9	9	10	8	9	10	9
			ΔTIC	7	8	9	8	8	7	9	8	9	7
			ΔTA	8	8	10	7	8	8	8	8	8	7
	colsample_bytree	{0.6, 0.8, 1}	ΔDIC	1	1	1	1	1	1	1	1	1	1
			ΔSIC	1	1	1	1	1	1	1	1	1	1
			ΔTIC	1	1	1	1	1	1	1	1	1	1
			ΔTA	1	1	1	1	1	1	1	1	1	1
	subsample	{0.6, 0.8, 1}	ΔDIC	1	1	1	0.6	1	0.6	0.6	0.6	1	0.6
			ΔSIC	0.8	1	0.6	0.6	0.8	0.8	0.6	0.8	1	0.8
			ΔTIC	1	1	1	1	1	1	1	0.8	1	0.8
			ΔTA	0.8	1	1	0.6	1	1	1	1	0.6	1

C.2 Predicted values

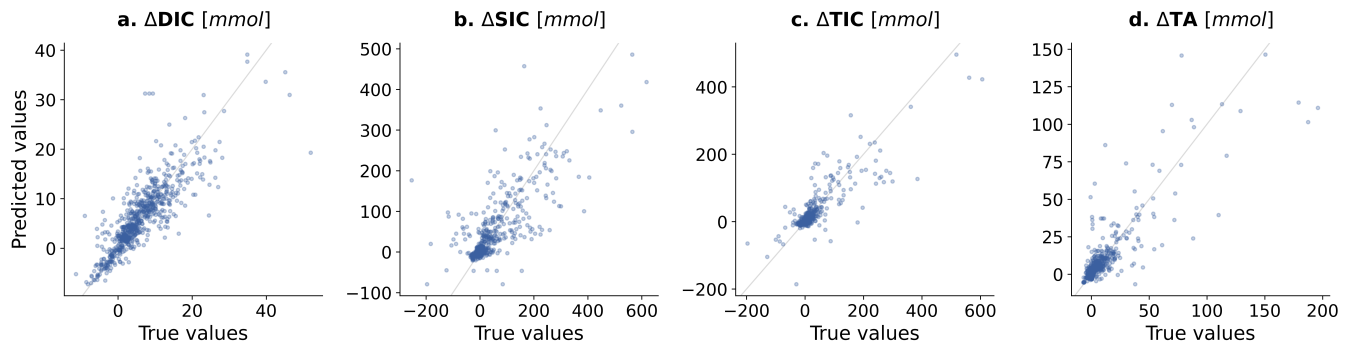


Figure C1: EXtreme Gradient Boosting predictions versus true values of the changes in amount of dissolved inorganic carbon (ΔDIC , a), solid inorganic carbon (ΔSIC , b), total inorganic carbon (ΔTIC , c) and total alkalinity (ΔTA , d).

C.3 SHapley Additive eXplanation results of the eXtreme Gradient Boosting models

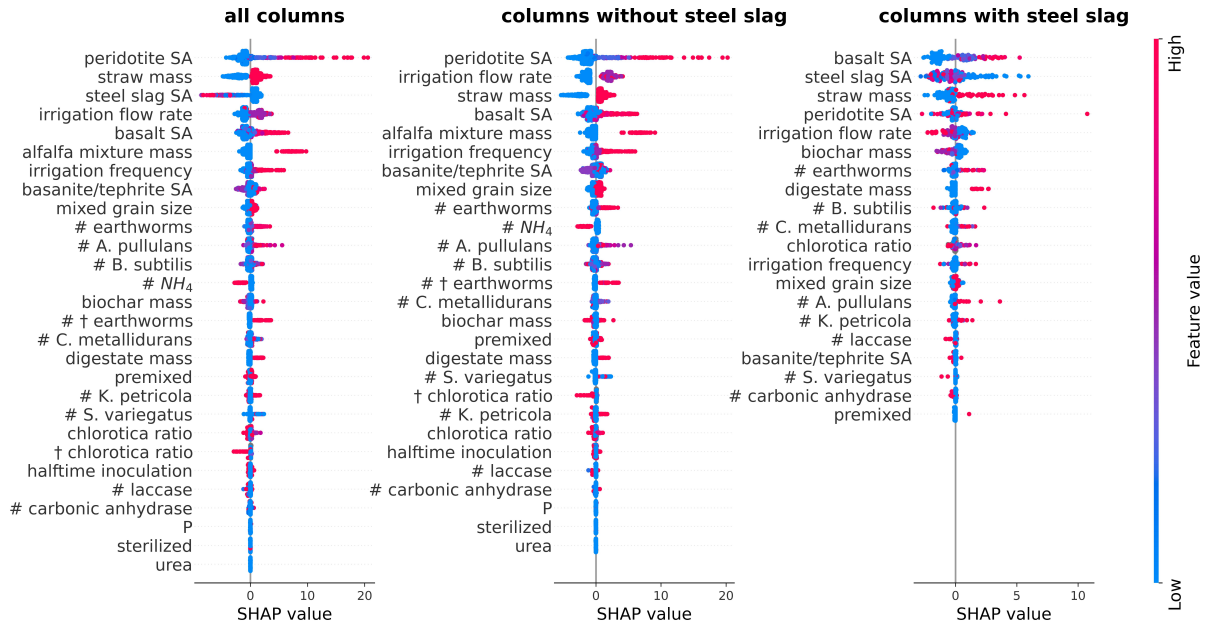


Figure C2: SHapley Additive eXplanation (SHAP) values of the varied features in the eXtreme Gradient Boosting prediction of the change in amount of dissolved inorganic carbon (ΔDIC). SA stands for surface area, # stands for added amount of biota.

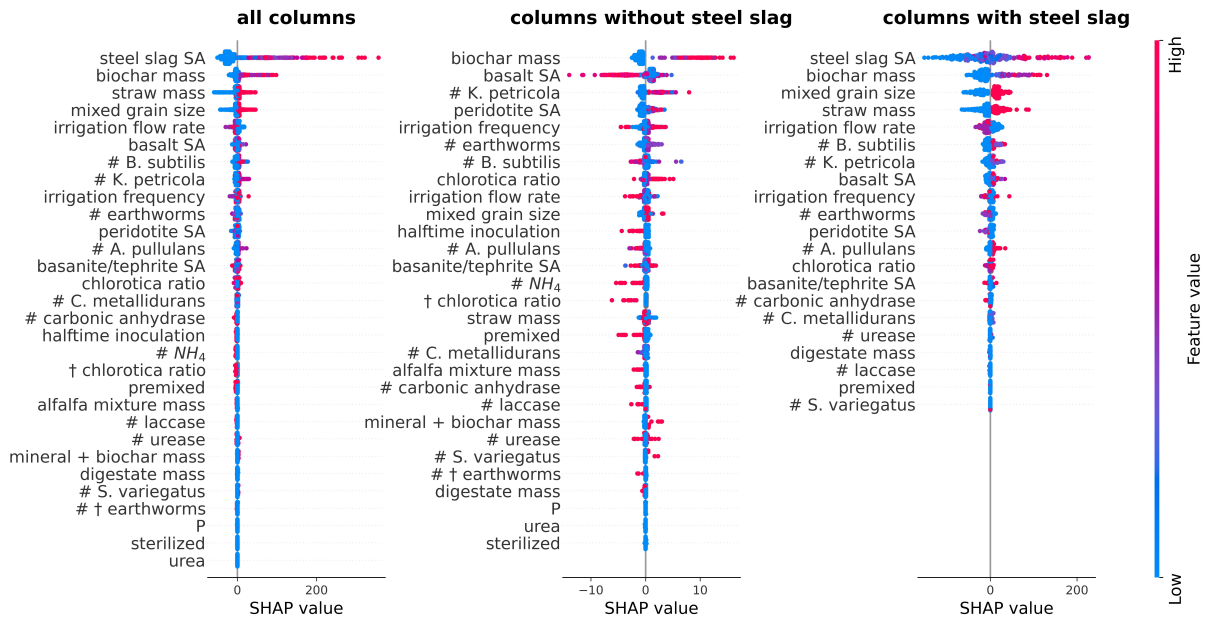


Figure C3: SHapley Additive eXplanation (SHAP) values of the varied features in the eXtreme Gradient Boosting prediction of the change in amount of solid inorganic carbon (ΔSIC). SA stands for surface area, # stands for added amount of biota.

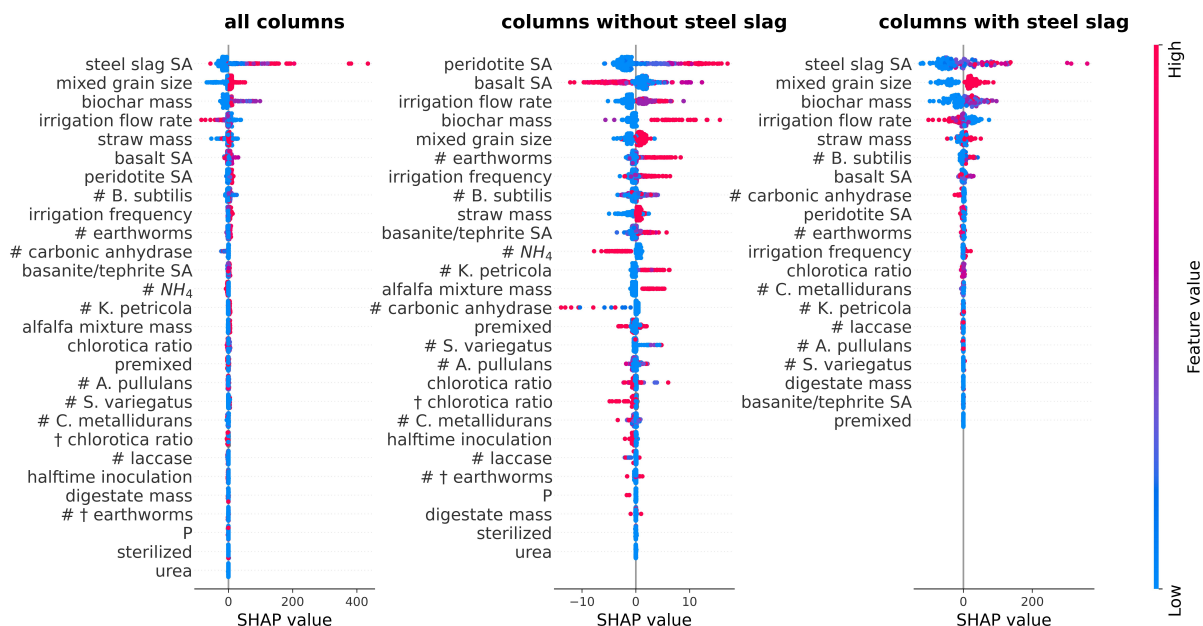


Figure C4: SHapley Additive eXplanation (SHAP) values of the varied features in the eXtreme Gradient Boosting prediction of the change in amount of total inorganic carbon (ΔTIC). SA stands for surface area, # stands for added amount of biota.

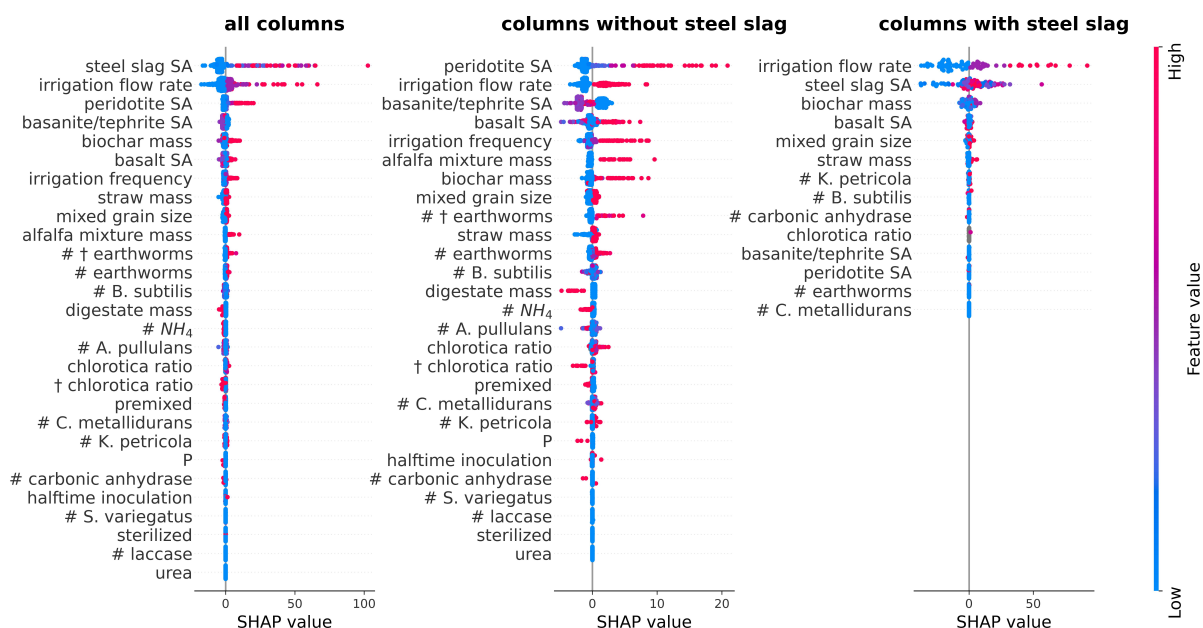


Figure C5: SHapley Additive eXplanation (SHAP) values of the varied features in the eXtreme Gradient Boosting prediction of the change in amount of total alkalinity (ΔTA). SA stands for surface area, # stands for added amount of biota.

C.4 Additional explainability results of the Least Absolute Shrinkage and Selection Operator and Random Forest models

C.4.1 Shapley Additive eXplanation results of the Random Forest models

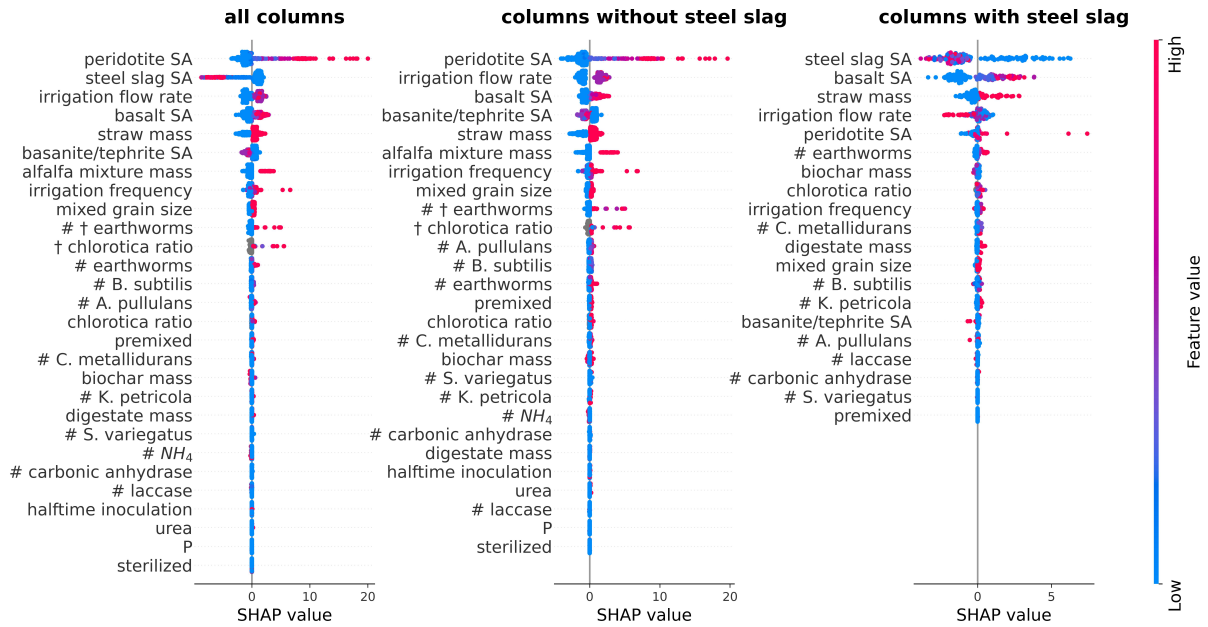


Figure C6: SHapley Additive eXplanation (SHAP) values of the varied features in the Random Forest prediction of the change in amount of dissolved inorganic carbon (ΔDIC). SA stands for surface area, # stands for added amount of biota.

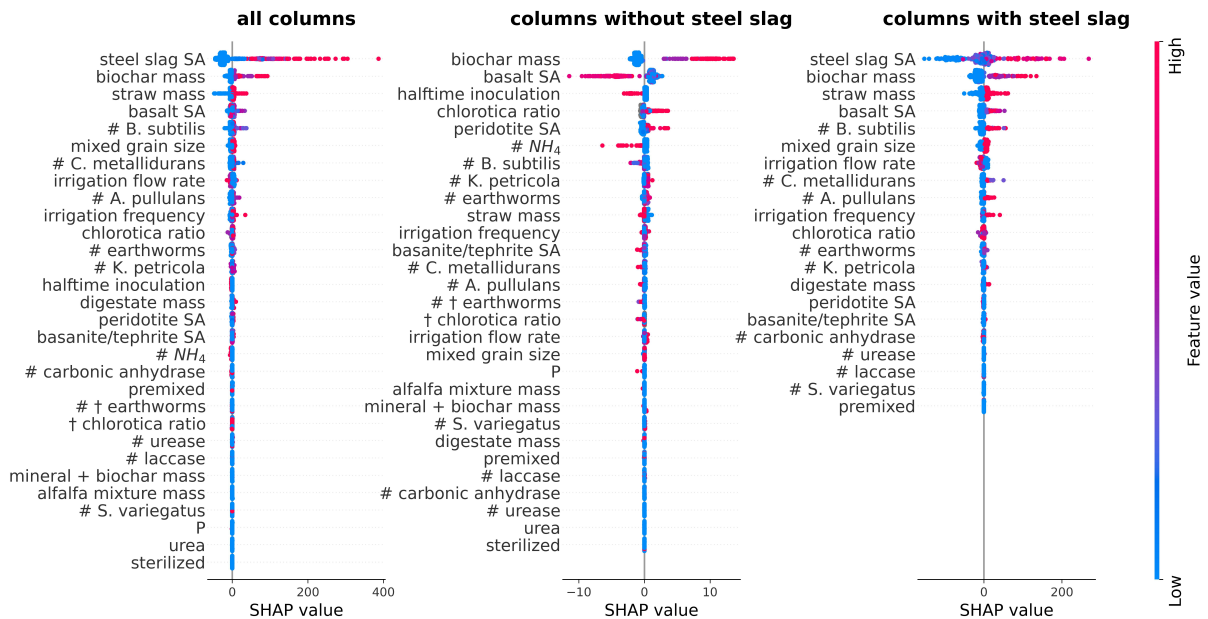


Figure C7: SHapley Additive eXplanation (SHAP) values of the varied features in the Random Forest prediction of the change in amount of solid inorganic carbon (ΔSIC). SA stands for surface area, # stands for added amount of biota.

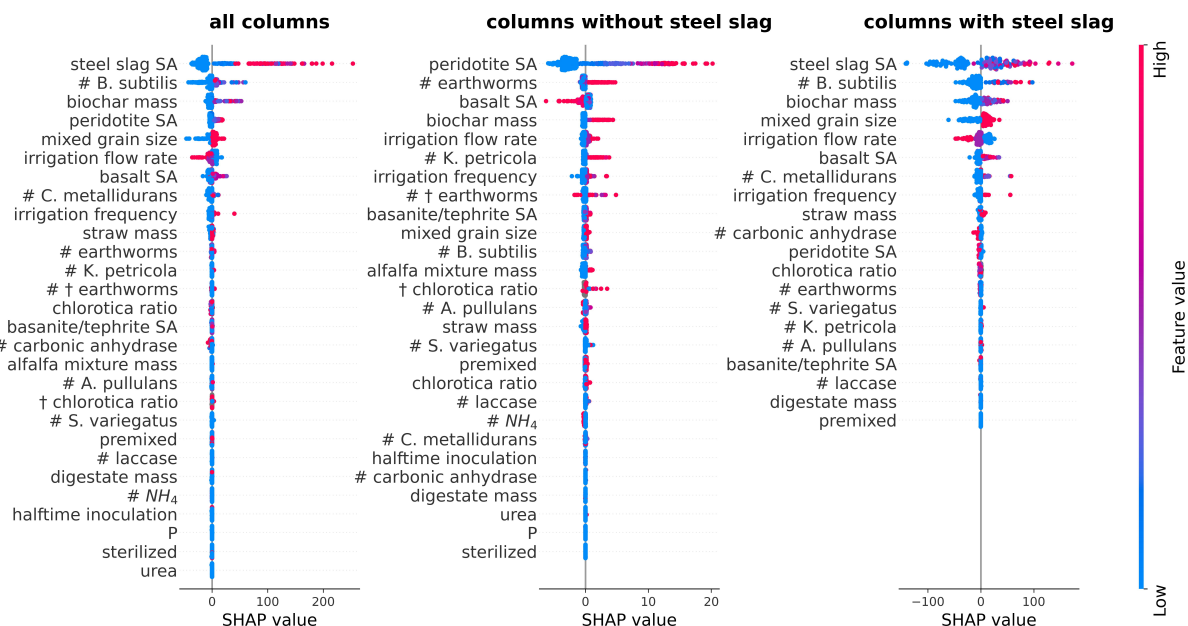


Figure C8: SHapley Additive eXplanation (SHAP) values of the varied features in the Random Forest prediction of the change in amount of total inorganic carbon (ΔTIC). SA stands for surface area, # stands for added amount of biota.

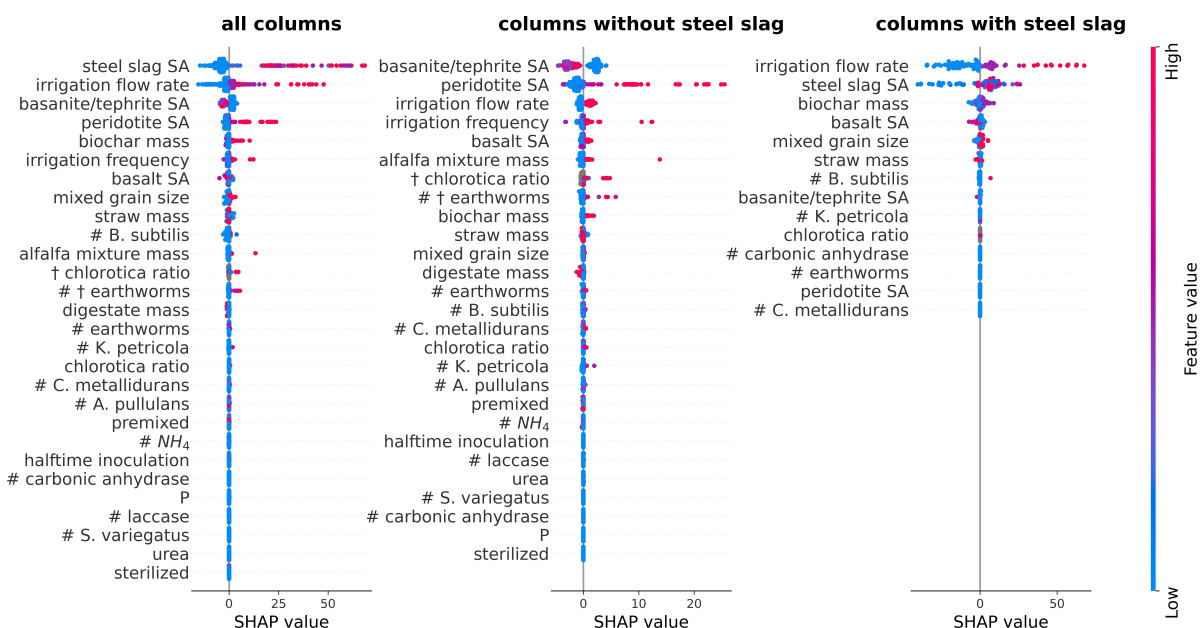


Figure C9: SHapley Additive eXplanation (SHAP) values of the varied features in the Random Forest prediction of the change in amount of total alkalinity (ΔTA). SA stands for surface area, # stands for added amount of biota.

C.4.2 Least Absolute Shrinkage and Selection Operator regularization results for change in amount of dissolved inorganic carbon

Table C2: Regression coefficients of the Lasso prediction of the change in amount of dissolved inorganic carbon. Depicted values are the averages and standard deviations of the coefficients of the ten models in the outer loop of the nested cross validation.

Feature	Coefficient	Feature	Coefficient
peridotite SA	18.35 ± 1.1	# K. petricola	0.67 ± 0.18
alfalfa mixture mass	8.96 ± 0.32	# C. metallidurans	0.29 ± 0.4
straw mass	5.09 ± 0.21	† chlorotica ratio	0.23 ± 0.41
basalt SA	4.43 ± 0.33	basanite/tephrite SA	0.2 ± 0.45
urea	3.7 ± 0.88	halftime inoculation	0.02 ± 0.06
digestate mass	2.32 ± 0.23	# urease	0.0 ± 0.0
# † earthworms	2.32 ± 0.47	mineral + biochar mass	0.0 ± 0.0
# S. variegatus	1.94 ± 0.45	chlorotica ratio	-0.01 ± 0.05
# A. pullulans	1.84 ± 0.29	# carbonic anhydrase	-0.09 ± 0.35
irrigation frequency	1.13 ± 0.08	P	-0.31 ± 0.4
# earthworms	1.03 ± 0.37	sterilized	-0.35 ± 0.25
# B. subtilis	0.9 ± 0.49	biochar mass	-0.5 ± 0.29
premixed	0.86 ± 0.18	# laccase	-1.18 ± 0.84
irrigation flow rate	0.81 ± 0.09	# NH_4	-1.49 ± 0.1
mixed grain size	0.78 ± 0.26	steel slag SA	-4.7 ± 0.41

C.5 Relation between organic and inorganic components of dissolved carbon and total alkalinity

Besides DIC and TA , also dissolved organic carbon (DOC) was measured in the leachates. Analogously to ΔDIC in section B.1.1, the change in amount of dissolved organic carbon (ΔDOC) can be calculated from the measured concentrations. Figure C10 illustrates the relations of ΔDIC , ΔDOC and their sum, ΔDC , with ΔTA . To investigate the contribution of ΔDIC and ΔDOC to ΔTA , linear regression was used to predict ΔTA based on ΔDOC and ΔDIC . This was done separately for (586) columns without steel slag and (139) columns with steel slag, and five fold CV was used. For columns with steel slag, the prediction had a low average accuracy with high variability across the folds ($R^2 = 0.28 \pm 0.38$; Fig. C11-b). In contrast, for columns without steel slag, the coefficient of determination (R^2) was high (0.90 ± 0.04 ; Fig. C11-a), and the resulting regression formula shows that TA is mainly inorganic:

$$\Delta TA [mmol] = (0.94 \pm 0.02) \Delta DIC [mmol] + (0.08 \pm 0.02) \Delta DOC [mmol] + (0.00 \pm 0.01) \quad (C.1)$$

Here, the mean and standard deviations are calculated from the coefficients of each of the five models in the cross validation.

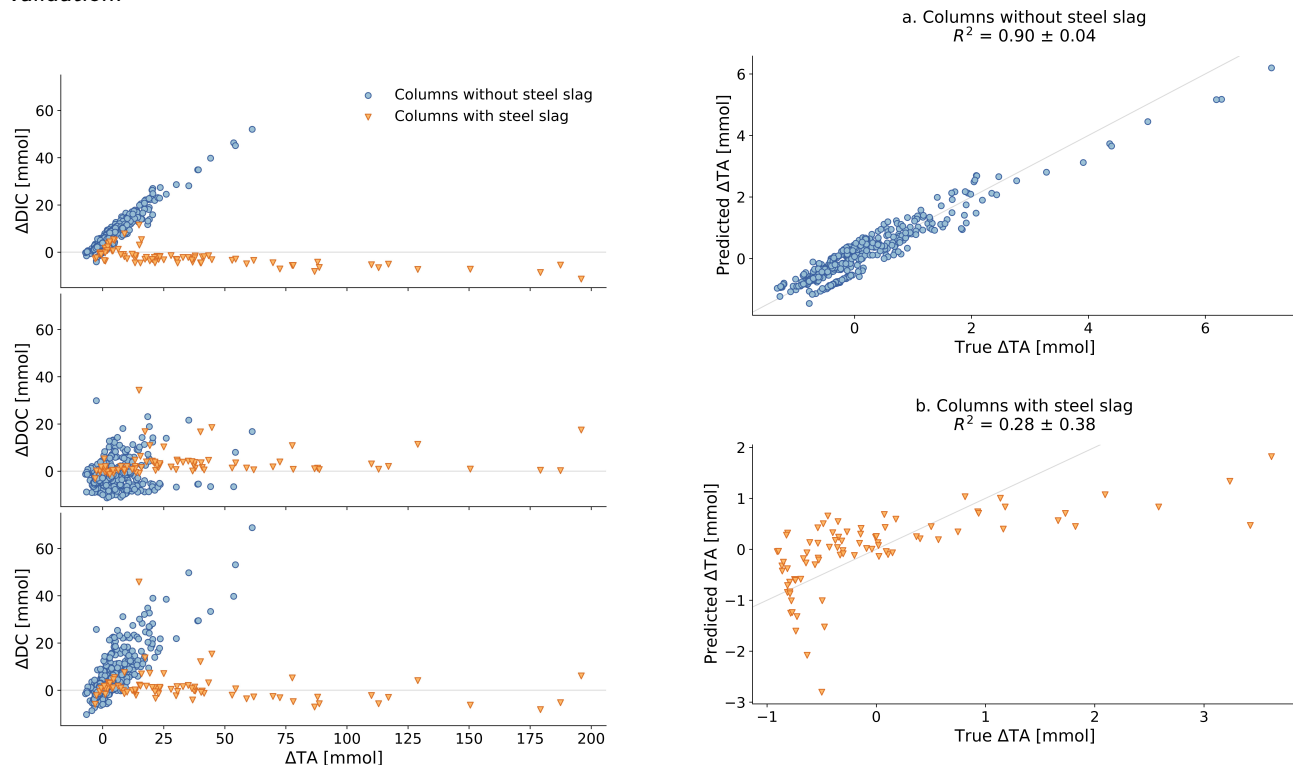


Figure C10: Changes in amounts of dissolved inorganic carbon (ΔDIC), dissolved organic carbon (ΔDOC) and their sum, dissolved carbon (ΔDC) as a function of the change in amount of total alkalinity (ΔTA)

Figure C11: Predicted change in amount of total alkalinity (ΔTA), estimated using linear regression based on changes in amounts of dissolved inorganic carbon (ΔDIC) and dissolved organic carbon (ΔDOC), plotted against ΔTA calculated from measured concentrations. The coefficient of determination (R^2) is reported as the mean $\pm$ standard deviation across the five folds of the nested cross validation.

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